

ASJID SIDDIQUE

Software Engineering Student | Full-Stack Developer | Building toward AI/ML Research

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SUMMARY

Fifth-semester BS Software Engineering student at **NUST** (CGPA **3.76/4.00**) with strong foundations in algorithms, operating systems, computer networks, databases, software engineering, and mathematics. Hands-on experience building machine learning, deep learning, and computer vision systems in Python and PyTorch, including independent work in defect detection and explainable fraud detection. Combines this with full-stack software engineering experience as founder and developer of a production e-commerce platform. Strong in independent learning and end-to-end implementation across data, modeling, deployment, and software engineering, with a growing focus on applied AI/ML research.

RESEARCH INTERESTS

Computer Vision • Deep Learning • Machine Learning • Explainable AI • Trustworthy AI • AI for Software Engineering

EDUCATION

National University of Sciences and Technology (NUST)

Sep 2024 – Jun 2028 (Expected)

BS Software Engineering | CGPA: **3.76 / 4.00**

Punjab College, Burewala

2022 – 2024

F.Sc. Pre-Engineering | Marks: 1083 / 1200

RESEARCH PROJECTS

PCBDefect-X — Deep Learning & Computer Vision [Live](#) | [GitHub](#) | [Research](#)

May 2026 – Present

Python • PyTorch • OpenCV • NumPy • Pandas • RF-DETR • ONNX Runtime • TensorRT • FastAPI • Next.js

- Developed an end-to-end **deep learning** and **computer vision** system for automated PCB defect detection and localization using the DeepPCB dataset and **RF-DETR Nano**.
- Built a reproducible data pipeline including deterministic dataset conversion, train/validation/test splitting, preprocessing, augmentation, and configuration management.
- Evaluated the model on a held-out test set of **366 images** across **6 defect classes**, achieving **78.8% mAP@50**, **39.3% mAP@50:95**, **90.0% precision**, **74.8% recall**, and **81.4% F1-score**.
- Developed inference and deployment pipelines using **ONNX Runtime** and **TensorRT FP16**, with parity validation and deployment smoke checks.

FraudShield — Machine Learning & Explainable AI [Live](#) | [GitHub](#) | [Research](#)

Jun – Aug 2026

Python • Scikit-learn • PyTorch • SHAP • Pandas • NumPy • FastAPI • Next.js

- Developed an end-to-end machine learning system for highly imbalanced fraud detection using **284,807 transactions** with approximately **0.17% fraud**.
- Implemented **train-only preprocessing** and **held-out evaluation** to prevent **data leakage**, with validation-based threshold optimization on an untouched final test set.
- Compared Logistic Regression, **Random Forest**, HistGradientBoosting, and PyTorch MLP models using ROC-AUC, PR-AUC, precision, recall, and F1-score.
- Selected Random Forest based on a PR-AUC of **0.803**, achieving **86.9% precision**, **76.8% recall**, and **81.6% F1-score** at the optimized threshold, and added **SHAP-based explainability** with a FastAPI inference service.

EXPERIENCE

Viro.pk — Founder & Full-Stack Developer [Live](#) | [GitHub](#)

Mar 2025 – Present

Next.js • React • Supabase • Tailwind CSS • PWA • REST APIs • Git • Vercel

- Founded, designed, built, deployed, and independently maintain a production e-commerce platform, managing its architecture, development, deployment, maintenance, and ongoing growth.
- Independently manage product discovery, inventory and order management, authentication, analytics, wishlist functionality, WhatsApp integration, COD checkout, and responsive PWA functionality.
- Designed the frontend and backend architecture and continuously iterate the platform through **40+ releases**, managing its full lifecycle including deployment, debugging, optimization, and release management.

SELECTED ENGINEERING PROJECTS

OS Kernel Simulator — Operating Systems & Full-Stack Engineering [Live](#) | [GitHub](#)

Feb – Apr 2026

Node.js • Express.js • React • MongoDB • JWT • REST APIs

- Built a full-stack operating-systems simulator implementing CPU scheduling (FCFS, SJF, Round Robin, Priority) and page-replacement algorithms (FIFO, LRU, Optimal).
- Implemented process synchronization concepts and backend APIs with authentication, input validation, pagination, and centralized error handling.
- Designed an MVC-oriented REST architecture with JWT authentication and responsive Gantt-chart visualization for simulation results.
- Developed interactive Gantt-chart visualizations and CSV-based simulation input; deployed frontend and backend via Vercel and Railway.

RELEVANT COURSEWORK

Computer Science & Engineering: Data Structures & Algorithms • Design & Analysis of Algorithms • Operating Systems • Computer Networks • Database Systems • Software Engineering • Software Construction • Software Design & Architecture • Object-Oriented Programming

Mathematics & Scientific Computing: Linear Algebra & ODEs • Calculus & Analytical Geometry • Probability & Statistics • Discrete Mathematics • Numerical Methods • Complex Variables & Transforms

Selected course topics: *Software Construction* — UML diagrams • Agile methodologies • software process/models • prototyping

Software Design & Architecture — system design concepts • architectural/design patterns • MVC • Proxy • software architecture concepts

Artificial Intelligence — search algorithms • intelligent agents • adversarial search

TECHNICAL SKILLS

Languages: Python • C++ • Java • JavaScript • TypeScript • MATLAB

Machine Learning & Explainable AI: Scikit-learn • Model Evaluation • Data Preprocessing • SHAP

Deep Learning: PyTorch • ANN • CNN • RNN/LSTM • Transformers

Computer Vision: OpenCV • Image Preprocessing • Data Augmentation • Object Detection

Data & Deployment: NumPy • Pandas • ONNX Runtime • TensorRT • FastAPI

Backend & APIs: Node.js • Express.js • REST APIs • JWT • bcrypt • MVC • Rate Limiting • Pagination

Frontend / Product: React • Next.js • HTML • CSS • Tailwind CSS • PWA

Databases: MongoDB • MySQL • PostgreSQL • SQL • Supabase • JDBC • Stored Procedures

Engineering / DevOps: Git • GitHub • Linux • Postman • Vercel • Railway